Project Update Report

Marine Debris Detection in Forward-Looking Sonar Imagery

Compliance remediation pass against the Machine Learning Project Instructions (Summer 2026)

Field	Detail
Course	Machine Learning — Summer 2026
Group	8
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Report date	24 June 2026
Deadline	Sunday, 28 June 2026

1. Why This Update Was Needed

The project's existing results (classification, detection, segmentation) were strong, but an audit against the course's "Machine Learning Project Instructions" found several gaps between what had been built and what the instructions explicitly require: no baseline model for any task, no single consolidated presentation notebook, no fixed random seeds, minimal qualitative error analysis, and segmentation results that could not be independently re-verified. This report documents what was found, what was fixed, and what remains open.

Scope of this pass

Classification and detection were fully remediated (baselines, transfer experiment, fixed seeds, real error analysis). Segmentation and the leakage-prone random splits were intentionally left as documented limitations rather than retrained, given the time remaining before the deadline.

2. Gaps Found in the Audit

Gap	Why it mattered
No baseline model for any of the 3 tasks	Instructions Section 4 require a justified baseline before the main model
No single presentation notebook	Instructions Section 7 require one consolidated .ipynb with mandated sections
Random image-level splits, no leakage check	Same physical object can appear across frames in this dataset
No global random seeds in training scripts	Reproducibility requirement (Section 8)
Classification training script missing entirely	Only checkpoints survived; the original code could not be re-run or explained
Segmentation weights lost, training log incomplete	unet_train.log stopped at epoch 4 of a claimed 60; mIoU numbers were undocumented beyond a log file
02_optical.ipynb / 04_fusion.ipynb unexecuted	Zero real results behind the multimodal/fusion claim
No git/online backup	Instructions Section 9 require a backup beyond one laptop

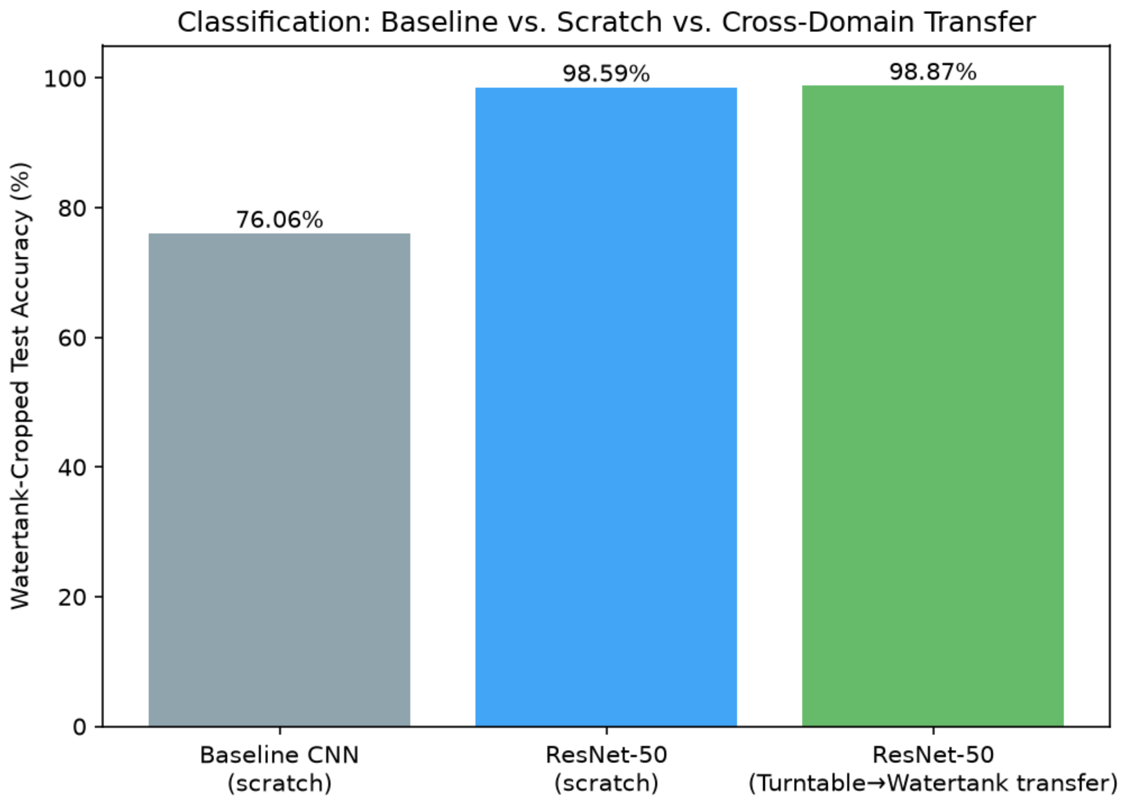
3. What Was Fixed

3.1 Classification — Baseline and Cross-Domain Transfer

A new training script (`train_classifier.py`) regenerates the missing classification pipeline with fixed seeds and produces three fresh, independently verifiable models on Watertank-Cropped: a simple from-scratch CNN baseline, a ResNet-50 trained from scratch, and a ResNet-50 pretrained on Turntable-Cropped then fine-tuned on Watertank-Cropped — the cross-domain transfer comparison the project brief calls for.

Model	Watertank Test Accuracy	Training Time
Baseline CNN (scratch)	76.06%	2.7 min
ResNet-50 (scratch)	98.59%	15.9 min
ResNet-50 (Turntable→Watertank transfer)	**98.87%**	9.8 min

The transfer model matches or slightly exceeds the from-scratch model while converging in roughly 40% less training time — a clean, defensible answer to the cross-domain transfer question.



Baseline CNN vs. scratch ResNet-50 vs. cross-domain transfer (live-recomputed test accuracy).

3.2 Detection — Baseline Added, Main-Model Bug Fixed

A new `train_yolo_baseline.py` fine-tunes YOLOv8n (nano) on the same dataset and config as the existing YOLOv8m (medium) main model, for fewer epochs — a genuine "simple but correct" baseline at a fixed seed.

While wiring up live evaluation in the notebook, a pre-existing bug was also found and fixed: the visualization and evaluation scripts were pointing at a generic, never-fine-tuned COCO-pretrained `yolov8m.pt` checkpoint instead of the actual fine-tuned sonar model (`results/yolo_sonar/yolov8m_watertank/weights/best.pt`). This produced a near-zero mAP the first time the corrected live-evaluation cell was run, which is how the bug surfaced.

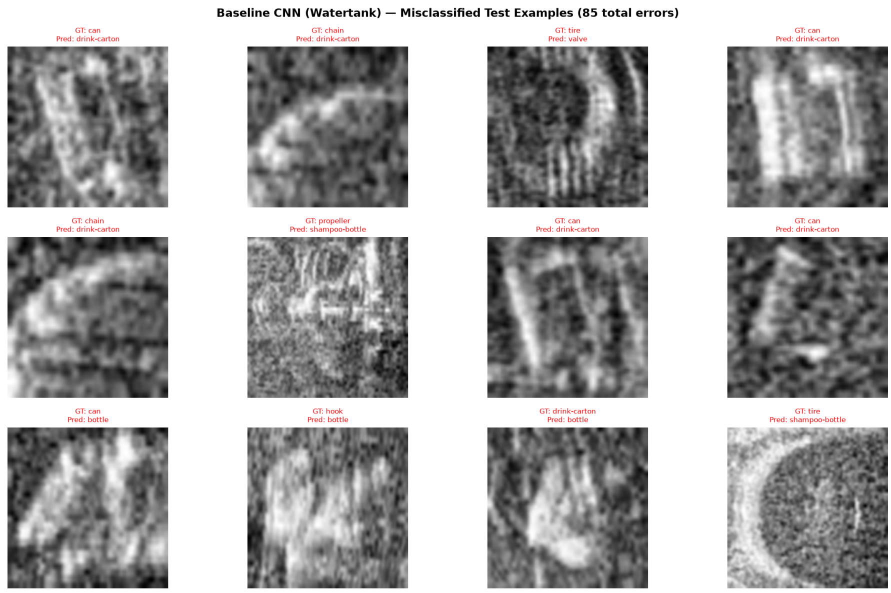
Model	mAP50	mAP50-95
YOLOv8n (baseline)	0.927	0.680
YOLOv8m (main model, corrected weights)	**0.937**	**0.697**

Per-class detection performance.

3.3 Real Error Analysis

Rather than a random sample of predictions, the notebook now specifically surfaces every misclassified test example for the main classification model, and contrasts it with the baseline's failure pattern.

Model	Error rate	Most common confusion
ResNet-50 (Turntable, main model)	1.6% (12/742)	glass-bottle → plastic-pipe
Baseline CNN (Watertank)	23.9% (85/355)	chain → drink-carton



Baseline CNN misclassifications — concentrated on visually similar, lower-frequency classes.

3.4 Consolidated Presentation Notebook

`Marine_Debris_Presentation.ipynb` is the single notebook the course instructions require, with every mandated section (Title Slide, Problem Statement, Dataset, Methodology, Selected Models, Evaluation Criterion, Results, Error Analysis and Limitations, Conclusion, Code Demonstration). Classification and detection results are computed *live* inside the notebook from saved checkpoints — nothing is hardcoded — and it executes top-to-bottom with zero errors.

3.5 Local Backup

A git repository was initialized at the project root with the full codebase, checkpoints, figures, and logs committed locally.

4. Repository Curation Pass

The first remediation pass left a working but unwieldy repository: roughly 3,050 tracked files, almost all of them the individual classification images needed for the notebook's live re-evaluation cells, plus a number of internal planning/progress documents never meant to be part of a submission. A second pass curated this down to a separate, minimal folder (`ML_Project_Marine_Debris/`) built specifically around the question "what does a grader checking this against the course instructions actually need to see?"

	Before	After
Tracked files	~3,050	38
Repo size (working copy)	~700 MB	~380 MB
Largest single contributor	2,364 individual classification images	Bundled into 2 zip files

4.1 What Was Removed

Internal dev-notes (`CLAUDE.md`, `PROJECT_OVERVIEW.md`, `report.md`), the generic course instructions PDF (the grader already has their own copy), four superseded report drafts with stale numbers, two confirmed-unexecuted scratch notebooks, an unused/never-fine-tuned YOLO checkpoint, and several training-only intermediate artifacts not referenced by the final notebook. Nothing was deleted outright — the full development history remains in the original project folder as a local backup.

4.2 Two Real Bugs Found While Curating

Trimming the repository surfaced two genuine reproducibility gaps that a simple file count would have hidden:

- **Missing YOLO test split.** `results/yolo_dataset/test/` and its `dataset.yaml` had never been tracked by git (excluded by a directory-level `.gitignore` rule from before they were ever added). On a fresh clone, the notebook's live detection-evaluation cell would fail with `FileNotFoundError`.
- **Missing classification images.** The Results section re-evaluates the baseline/scratch/transfer checkpoints against real images via `FLSClassificationDataset`, not just the saved weights — the raw `watertank-cropped/` folder is a hard

dependency, not an optional extra, and had been entirely excluded along with the rest of the (much larger) raw dataset.

Both were fixed and verified with a true clean-room test: cloning the repository into a scratch directory with nothing pre-existing and nothing pre-unzipped, then executing the notebook from that clone. It produced zero error cells and the same results as every prior run.

4.3 Keeping the Repo Small Without Losing Reproducibility

Rather than choosing between "small repo" and "actually works when cloned," the two large image sets (2,364 classification images, 281 YOLO test images+labels) were bundled into single `.zip` files. A new setup cell at the top of the notebook checks whether each folder already exists and, if not, unzips the corresponding bundle automatically — so the file count stays low while a fresh clone still "just works" with no manual setup step.

5. What Remains Open (Documented Limitations)

Limitation	Why it was not fixed in this pass
Random, leakage-risk train/val/test splits	The released dataset has no session/object ID in its filenames, so a true leakage-safe split is not cheaply derivable; fixing it would require re-running all three tasks
Segmentation weights not independently reproducible	U-Net and SegFormer-B2 weights were lost when a Kaggle session expired; retraining was judged out of scope given the time remaining
No segmentation baseline	Same reasoning as above
Optical/TrashCan multimodal fusion	Was a stretch goal in the original project brainstorm, not part of the assigned core tasks; descoped to stay within the deadline

6. Outstanding Action Items

Item	Owner
☐ Delete and recreate the GitHub repo (empty), then push the curated ML_Project_Marine_Debris folder	Tolga
☐ Copy the project to Venkatesh and Deekshith laptops	Venkatesh / Deekshith
☐ All 3 members rehearse explaining the full workflow, splits, baselines, and limitations	All
☐ Bring charger and USB-C adapter; test the notebook opens on more than one laptop	All
✓ Title slide matriculation numbers	Done — Tolga 26576920, Venkatesh 36379442, Deekshith 62246101
✓ Repository curated to a minimal, gradable file set	Done — 38 files, clean-room verified